

Dengue, Yap Main Islands, Micronesia, 2011

Overview

This dataset describes a dengue fever outbreak on the Yap Main Islands, Federated States of Micronesia, in 2011. Dengue serotype 2 caused 978 cases among 7,391 residents (13% attack rate). This outbreak occurred concurrently with the Fais Island outbreak but in a larger population.

Source: Funk S, Kucharski AJ, Camacho A, Eggo RM, Yakob L, Murray LM, Edmunds WJ (2016). Comparative analysis of dengue and Zika outbreaks reveals differences by setting and virus. *PLOS Neglected Tropical Diseases* 10(12):e0005173.

Loading the data

```
using DataFrames
using Outbreaks
df = Outbreaks.dengue_yap_2011()
size(df)
```

(185, 3)

First few rows

```
first(df, 6)
```

	onset_date	nr	value
	Date	Int64	Int64
1	2011-07-07	7	0
2	2011-07-14	14	0
3	2011-07-21	21	0
4	2011-07-28	28	0
5	2011-08-04	35	0
6	2011-08-11	42	0

Summary statistics

```
describe(df)
```

	variable	mean	min	median	max	nmissing	eltype
	Symbol	Union...	Any	Any	Any	Int64	DataType
1	onset_date		2011-07-07	2011-11-24	2012-04-12	0	Date
2	nr	147.0	7	147.0	287	0	Int64
3	value	5.28649	0	4.0	29	0	Int64

Epidemic curve

```
sort!(df, :onset_date)
println("Onset range: $(minimum(df.onset_date)) to $(maximum(df.onset_date))")
println("Total value: $(sum(df.value))")
for row in eachrow(first(df, 30))
```

```

    if row.value > 0
      bar = repeat("█", min(row.value, 60))
      println("${row.onset_date} (wk ${row.nr}) | $bar ${row.value}")
    end
  end
println("... (${nrow(df)} rows total)")

```

Onset range: 2011-07-07 to 2012-04-12

Total value: 978

```

2011-09-01 (wk 63) | █ 1
2011-09-02 (wk 64) | █ 2
2011-09-09 (wk 71) | █ 1
2011-09-15 (wk 77) | █ 1
2011-09-16 (wk 78) | █ 2
2011-09-19 (wk 81) | █ 1
2011-09-22 (wk 84) | █ 2
... (185 rows total)

```